Brendan Clark

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Education

Rutgers University — New Brunswick, NJ
Ph.D. Atmospheric Science, Alan Robock's group — 2025

University of Massachusetts — Amherst, MA
B.S. Environmental Science, certificate in GIS — 2020

Experience

Cornell University — Ithaca, NY
Research Associate — 2025-present

- Integrating eddy covariance measurements of photosynthesis, surface aerosol observations, radiative transfer modelling, and satellite imagery to develop statistical models quantifying the radiative effects of wildfire smoke and its impacts on crop and forest growth.
- Collaborated with **CarbonPlan, Inc.** to create a Python-based pipeline, managed using Git, for statistical downscaling of climate model simulations and built tools that enable public access to downscaled data through AWS cloud infrastructure.
- Developed machine learning models in Python using Scikit-learn and Statsmodels to assess the impacts of climate change on forest carbon uptake and storage and compared model predictions with numerical simulations to identify limitations in current Earth system models.

Rutgers University — New Brunswick, NJ
Research Assistant — 2020-2025

- Utilized FORTRAN, Linux, shell scripting, text-processing tools, and job scheduling to run simulations and develop code for the Community Earth System Model (CESM) on the Derecho supercomputer.
- Downscaled and bias-corrected climate model output and conducted global crop model simulations to assess the impacts of climate change and potential climate interventions on future crop production and nutritional quality.

University of Massachusetts — Amherst, MA
Research Intern — 2019-2020

- Analyzed trace gas emissions from shale samples mixed with water using gas chromatography to investigate secondary methane and nitrous oxide emissions associated with hydraulic fracturing.

Massachusetts Department of Conservation and Recreation — Amherst, MA
GIS Data Analyst Intern — 2018-2019

- Digitized and integrated property boundary datasets into a GIS platform, enabling visualization of property line data through a web-based interface.

Skills

Languages: Python (Pandas, Scikit-learn, Xarray, NumPy, Matplotlib), R, MATLAB, FORTRAN
Abilities: Scientific writing, creative problem-solving, collaboration, and project coordination.